

FULL METAL RACKET

Twelve channels that behave like one kit in one room.



An analogue drum machine with twelve voices, a thirty-two step sequencer and two hundred kits. Nothing in it is sampled and nothing in it is a clone: every voice is built from the same handful of mechanisms, which is why the machine sounds like one machine rather than twelve circuits sharing a jack.

The thesis, in one paragraph

A drum machine is usually twelve independent synthesisers wired to a mixer. The parts never quite belong together, and everyone reaches for bus compression to fake the missing glue. This one is built the other way round. Hit the snare and the toms ring, because **THE WEB** lets any trigger excite the other channels' resonators. Everything feeds one shared shell, **KIT BODY**. Everything shares one power rail, so a hard kick dips the whole machine in level *and pitch* for a few milliseconds — which is the part a compressor cannot imitate. And **AGE** gives every channel its own component tolerances, drifting slowly, so no two hits are quite identical.

At zero, every one of those is exactly absent. The machine will render bit-identically with the glue switched off, so you can hear precisely what each one is doing.

What it is made of

Twelve voices	Two kicks, two snares, three toms, a percussion wildcard, closed and open hats, two cymbals — in the order every classic machine uses, read from the floor upward.
No VCA	On the ping models <i>DECAY</i> is the resonator's Q, the way the 808 did it. There is no amplitude envelope shaping a tone; the circuit rings and stops ringing.
Sequencer	32 steps, 16 patterns, per-lane length, division, direction and swing. Derived from the host's bar position, not counted, so looping and relocating are free.
200 kits	Generated from a seed, filed by category, each with a name. The same number is the same kit on any machine, forever.
MIDI first	The sequencer is optional. Play it from a pad controller and it never argues with your DAW.
Thirteen outputs	Main stereo plus a mono aux per channel, all off by default until a host asks for them.

There is no ACCENT channel. Every classic machine has one, first on the selector, and you will look for it. Accent here is *per step* — a velocity on each step of each lane — which is the same idea in its modern form and rather more use.

1 Put it on a MIDI track.

It answers the General MIDI drum map, and also C3–B3 for the twelve channels in panel order — whichever your controller sends.

2 Turn the kit dial.

The arrows either side of the display step through two hundred kits; clicking the display opens the browser, where they are filed by category. RANDOM on the left rail throws you somewhere at random.

3 Press STEPS and draw something.

Click a cell to turn it on. Press the ► key and it runs, on your DAW's tempo if the transport is rolling and on its own if not.

4 Turn RAIL SAG up.

Then play the kick hard. That is the machine breathing — one supply, shared by twelve voices.

5 Turn REBOUND up on the snare.

A little is a flam, more is a drag, a lot is a buzz roll. One knob for three techniques that are otherwise miserable to program.



The VOICES page. Twelve strips, every knob visible at once — the tactile wall is most of the point. The rail on the left switches pages and holds the things you reach for while playing; MASTER is the strip on the right.

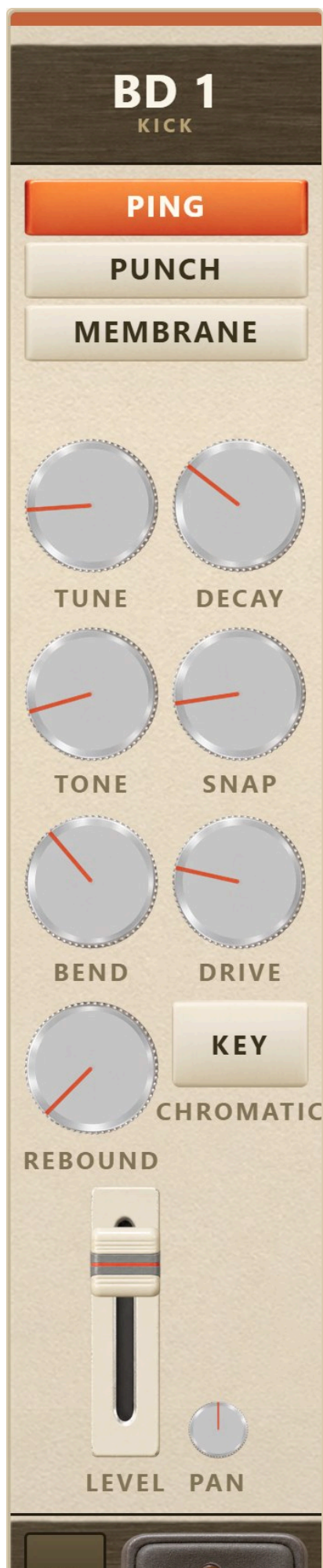
Kick, snare, toms low to high, percussion, then metal. Two conventions are near-universal in the tradition and both are kept: the toms run low to high left to right, and the percussion voice sits between the toms and the metal — which is exactly where the 808 put its claves and cowbell.

CHANNEL	MODELS	WHAT IT IS
BD 1 · BD 2	PING / PUNCH / MEMBRANE	Two kicks so you can have a long one and a short one, or a body and a click, without robbing another voice.
SD 1 · SD 2	SHELL / WIRES / CLAP	CLAP is a snare model rather than its own channel — a clap is a snare drum played by a crowd.
TOM 1–3	MEMBRANE / CONGA / BLOCK	Three is the classic count; the 808 and the 909 both had three.
FX	BELL / WOOD / ZAP / VOX	The wildcard, in the tradition's percussion slot. WOOD and BELL <i>are</i> the claves and the cowbell.
CH · OH	CRASH / RIDE / SPLASH	Closed and open hat, shipped LINKed — see below.
CY 1 · CY 2	CRASH / RIDE / SPLASH	Multi-band decay, so the top goes before the body does, the way a real cymbal dies.

The hat is two channels that behave like one

A hi-hat is not one instrument, it is a *pair of cymbals with a pedal*, and its whole musical function is that the open one is cut off by the closed one. Two channels buy four things a single dual-trigger circuit could not: its own sequencer lane — decisive, you want a CH lane and an OH lane rather than an openness flag per step — its own aux out, its own level and its own pan.

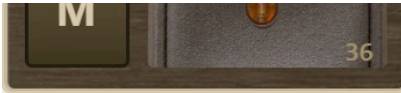
What that risks losing is that they are one pair of cymbals. So the pair ships **LINKed** (the button on the left rail): OH mirrors CH's tune and model, and CH chokes OH hard. Unlink and they are independent, which is occasionally wanted and always physically wrong — so link is the default, because the correct thing should be the easy thing. When linked, OH's TUNE and MODEL show and drive the pair, and are ringed to say so.



The six knobs

TUNE	The resonator's frequency. On the metal channels it also moves the band filters, because that is where their pitch actually lives.
DECAY	On the ping models this <i>is</i> the Q. There is no VCA: the circuit rings for as long as it rings.
TONE	The balance of body against noise, and where the noise sits.
SNAP	The attack transient — the stick, before the drum has decided what note it is.
BEND	The pitch envelope. Every good kick is a falling note.
DRIVE	Saturation into the channel's own limiter.

Underneath, two mechanisms work without a knob: hitting harder makes the drumhead *tense*, so a hard hit starts sharp and settles; and the resonator's damping depends on amplitude, so a loud hit does not simply sound like a quiet one turned up.



One strip: model, six knobs, REBOUND, KEY, level, pan, mute and the trigger pad with its MIDI note.

REBOUND, and KEY

REBOUND turns one trigger into a modelled bounce: shortening intervals, falling velocity, the way a stick actually behaves on a head. Low is a flam, middle a drag, high a buzz roll.

KEY makes the channel chromatic. MIDI notes track the resonator properly, and the decay scales with pitch so a low note rings longer, like a real one. A tuned kick is the most-used trick in modern production and very few drum machines do it correctly. The exact drum map is consulted first, so a channel in KEY mode never steals another channel's note.

Five controls sit above the strips and none of them belongs to a channel. They are what makes the twelve behave like one kit, and each of them is exactly absent at zero.

RAIL SAG	One power supply, shared. A hard hit pulls it down and everything dips for a few milliseconds — level and <i>pitch</i> . The pitch component is why sag glues in a way a compressor cannot: a compressor can duck the level, but it cannot make the whole kit sag flat for an instant and spring back.
KIT BODY	One resonator that every channel feeds, with its own tuning and decay. The shell the machine sits in. Not a reverb — the world rack has a reverb — a coupling.
BLEED	THE WEB. Any trigger <i>excites</i> the other channels' resonators without firing their envelopes, so a snare rings the toms and a kick buzzes the hats. This is the thing bus compression is trying to fake, and it costs almost nothing because the resonators are already there.
AGE	Per-channel component tolerances that drift slowly, so no two instances are identical and no two hits are. At zero, determinism is bit-exact.
KIT TUNE	Transposes the whole machine two octaves, metal included.

KIT MORPH, and the A / B slots

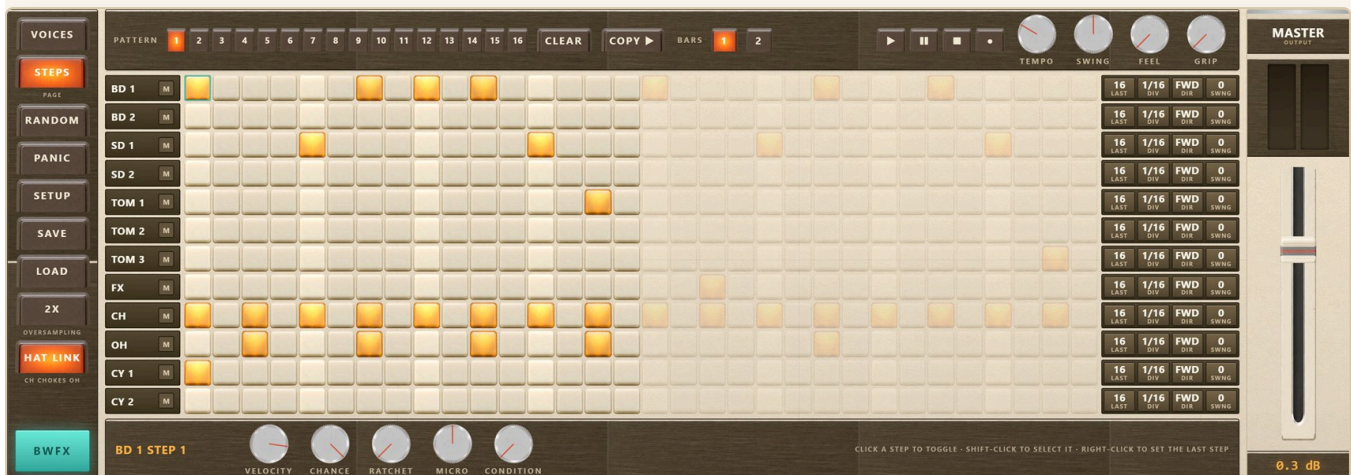
A and B are *slots*, not switches. Press **A** and it holds whatever the machine is doing at that moment; change the machine as much as you like; press **B**. The MORPH knob then blends between the two — every continuous parameter interpolated, every switched one flipping at its own threshold along the way, so the kit changes character rather than cross-fading.

MORPH is dim until both slots hold something, because until then there is nothing to morph toward. Pressing a lit slot again re-captures it; shift-clicking empties it.

The morph moves the panel too. It is applied to the engine's copy of the parameters and never written back to the knobs — a fader that silently rewrote a hundred controls would wreck both its own automation and theirs. But the panel computes the same blend for display, so the knobs sweep as you move the fader even though nothing has been overwritten.

THE HAND

On the STEPS page, **FEEL** and **GRIP** humanise the playing — and not by adding random milliseconds, which never sounds human. A drummer has limbs, and a limb cannot play two things at once: the second hit displaces, which is where flams come from by themselves. Add a little push and pull per subdivision and some velocity carried over from the previous hit on the same limb, and the machine breathes instead of stuttering.



Thirty-two steps, twelve lanes. The dimmed columns are past the lane's LAST step — they hold what you drew, they simply are not in the loop. Each lane's LAST, DIV, DIR and SWNG sit at the right-hand end of its row.

How it keeps time

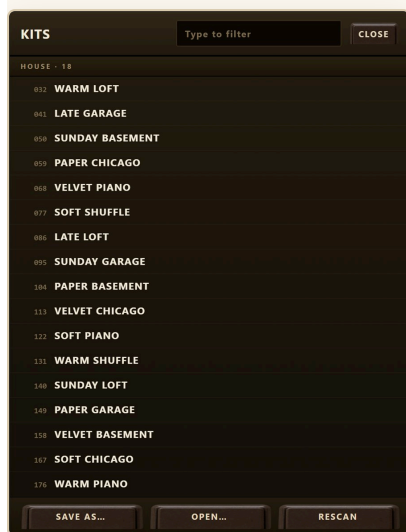
Every step is *derived* from the host's bar position rather than counted forward. That is not a detail: it means looping, relocating and tempo changes are free and exact, and a bounce matches what you heard. Measured against the host clock, the worst placement error is one sample. With no transport rolling it free-runs on the TEMPO knob, and the moment your DAW starts, the DAW takes over.

The transport

- ▶ RUN Starts from step one.
- ⏸ PAUSE Stops where it is; press again to continue from there. SPACE does this too. Two keys rather than one toggle, because "start again" and "carry on" are different intentions and a single button has to guess which you meant.
- STOP Stops and rewinds to step one.
- RECORD Writes what you play — pads or MIDI — into the current pattern, at the *nearest* step of that lane. Not the step just passed: quantising backwards makes everything you played sound late.

Per lane, and per step

LAST	Where that lane's loop ends, 1 to 32. Set the hats to 16 and the toms to 12 and they drift against each other for four bars before they agree again. Right-click a step to set LAST there.
DIV	The lane's clock division, so one lane can run at half or double speed.
DIR	Forward, reverse, pendulum or random.
SWNG	Swing <i>per lane</i> , which is how grooves actually work — swung hats over a straight kick.
SHIFT-CLICK A STEP TO SELECT IT, THEN USE THE FIVE KNOBS BELOW	
VELOCITY	The accent, per step. Fifteen decibels of range, not five.
CHANCE	How often that step actually fires.
RATCHET	Two to eight hits in the space of one, with their own ramp.
MICRO	Nudge the step off the grid, ahead or behind.
CONDITION	Play every other pass, every fourth, on fills, off fills, or only if the previous conditional step played.
PATTERN chooses one of sixteen; BARS stretches the thirty-two steps over one bar or two; COPY takes the current pattern somewhere else and CLEAR empties it.	



Two hundred kits, filed by category and filterable.

The two hundred

Each kit is generated from its number, so the same number is the same kit on any machine and always will be. The category is decided *first* and the parameters are then made to fit it — which is why a kit filed under TECHNO sounds like techno rather than being labelled after the fact. The names are built from grids that share no nouns between categories, so all two hundred are unique.

The arrows either side of the display step through them; clicking the display opens the browser; RANDOM dials one at random.

Your own patches

SAVE writes a `.fmrkit` file — the whole machine, sequencer and world rack included. LOAD opens one. A loaded patch names itself on the display in red, marked **USER PATCH**, and dialling a kit takes it back.

Patches live in `Documents\Brokild patches\Full Metal Racket`, where every Brokild plugin has a folder of its own — deliberately not beside the installed `.vst3`, which is somewhere an installer can reach and replace.

Loading always starts at step one. With a host rolling, every step is derived from the bar so this was already true; free-running, the internal clock used to keep whatever phase it had. Loading a patch, dialling a kit, RUN and STOP all put it back to the top.

The world rack

The teal globe at the foot of the rail opens **Brokild World FX** — the same rack of effects and the same SPECTRA characters that live in every Brokild instrument. A rack built here will open in any of them. It is empty by default and adds nothing until you put something in it.

The SPECTRA characters reach further than the effects do: they possess the machine itself, detuning, panning and sagging the twelve voices from the inside rather than treating the mix.

A kit in one room

Any kit. **BLEED** to about 40, **KIT BODY** to 30, **RAIL SAG** to 50. Play a kick and snare pattern with nothing else.

The toms you are not playing start answering the snare, and the whole machine dips under the kick. This is the sound bus compression is an apology for.

A bassline out of the kick

BD 1 to the **MEMBRANE** model, **KEY** on, **DECAY** long, **BEND** low. Play notes an octave below middle C.

A tuned kick that holds pitch and rings longer down low, because the decay scales with the note.

The polyrhythm

Hats **LAST** 16, toms **LAST** 12, kick **LAST** 32. Draw something simple in each.

Three loops of different lengths that agree only once every four bars. Nothing else on the panel needs touching.

A drummer, not a grid

A busy hat pattern. **GRIP** to 60, **FEEL** to 40, per-lane **SWNG** on the hats only.

Flams appear where two limbs collide, the hats lean forward, and none of it is random jitter.

One fader, two kits

Dial a kit, press **A**. Dial a very different one, press **B**. Automate **MORPH** across eight bars.

The kit changes character rather than cross-fading, because the switched parameters flip at staggered points along the way.

The buzz roll

SD 1, **REBOUND** up to about 70, one step per bar. Then try 20.

At 70 a press roll; at 20 a flam. One knob, and the intervals shorten the way a real stick does.

1

Unzip.

Copy the whole `Full Metal Racket.vst3` folder — not just the file inside it — into `C:\Program Files\Common Files\VST3\`. A sub-folder such as `...\VST3\Brokild\` is fine; hosts look inside. Windows will ask for administrator permission, which is normal.

2

Rescan.

In Ableton Live: Preferences → Plug-Ins → Rescan. It appears as an instrument called **Full Metal Racket**, by Brokild.

3

Or just run it.

`Full Metal Racket.exe` in the same zip is the standalone. No installation needed.

Settings

SETUP on the rail holds the things that belong to your machine rather than to the patch: the button finish (black or metal), and the panel light. Both are saved on this computer and not in the project, so a session always sounds the way it was saved even if it looks different.

The panel light is a gamma curve rather than a brightness control — it lifts the shadows, where everything on a panel this pale actually lives, and leaves the highlights alone instead of flattening them.

Outputs and oversampling

Main stereo, plus one mono aux per channel, all disabled until the host enables them. The **2X** button on the rail sets oversampling: 1× is about six percent of a core, 2× ten, 4× eighteen. Offline renders are not forced to a higher rate — what you hear is what you get.

What the bench proves

Every build is run through an offline bench that renders real audio and measures it: tuning within half a cent, a played octave at 2.0004, fifteen decibels of velocity range, sequencer placement inside one sample of the host clock, every one of the two hundred kits bounded and audible and unlike every other, and no channel ringing past its own decay. Sixteen hundred and ten checks at the time of writing. It is run after every change to the sound, and the numbers in this manual come from it rather than from anybody's opinion.

If the window comes up blank, install the free Microsoft Edge WebView2 Runtime and reopen the plugin. The interface is drawn in a WebView2 surface, which every current Windows 10 and 11 already has. The audio keeps working with the window shut either way.

measured crop regions. Free: no installer, no account, no telemetry, no network access at all.

Peter Boggild / Brokild, August 2026.